

Segmentation Network for Agricultural Weed Detection

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Abstract—Semantic segmentation of crops and weeds enables targeted herbicide application, reducing manual labour, chemical costs and environmental impact. A key challenge is achieving reliable segmentation from small, class-imbalanced datasets. This paper presents two U-Net segmentation models trained on a 50-image subset of the SugarBeets 2016 dataset. The baseline model establishes a reference performance under standard training conditions. The improved model addresses data scarcity and class imbalance through data augmentation, class-focused patch extraction, weed-dominant image oversampling, and iteratively refined class weights informed by confusion matrix analysis. The improved model achieves a mean IoU of 0.78 and mean F1 of 0.87, compared to 0.51 and 0.62 for the baseline, with weed IoU improving by 237% and crop IoU by 91%. Results demonstrate that these training strategies substantially improved minority class segmentation performance.

Index Terms—semantic segmentation, weed detection, U-net, data augmentation, patch extraction

I. INTRODUCTION

Agricultural weed detection is a growing computer vision problem. Manual weed detection is labour intensive and time consuming, it also does not scale to larger fields. Without reliable weed detection, herbicide application becomes non-targeted, where the whole field is sprayed rather than just the weed locations. This increases chemical costs and has significant negative environmental impacts. Robots or drones equipped with cameras could offer a scalable alternative but require accurate per-pixel segmentation to enable spatially precise herbicide application. A key challenge in this domain is the scarcity of labelled training data, as annotation of agricultural imagery is costly and time consuming. This paper addresses this challenge by training and evaluating two U-Net segmentation models on a dataset of only 50 images. The baseline model establishes a reference performance under standard training conditions. The improved model incorporates data augmentation and class-focused patch extraction to address the data scarcity and class imbalance.

II. METHODOLOGY

This project was implemented entirely in MATLAB R2025b using the Computer Vision Toolbox and Deep Learning Toolbox. Training was performed on CPU only, as MATLAB does not support GPU acceleration on macOS. All decisions related to network architecture & depth, input resolution, batch size, and number of epochs were made under this constraint.

A. Dataset and Preprocessing

The dataset used in this project is the SugarBeets 2016 dataset [1], collected by an agricultural field robot on a sugar beet farm near Bonn, Germany. Each RGB image has a resolution of 1296×966 pixels with corresponding pixel-level segmentation groundtruth encoding three classes using a colour map: background (black), crops (green), and weeds (red).

An exploratory data analysis revealed a major class imbalance. The background accounts for 94.5% of the pixels; weeds 3.8% and crops 1.6%. This imbalance directly informed the loss function design and patch extraction strategy described in later subsections.

Images were resized from 1296×966 to 256×320 prior to training. This target size preserves the original 4 : 3 aspect ratio, avoiding geometric distortion that would arise from square resizing. Pixel values were normalised to $[0, 1]$ to stabilise gradient descent during training.

The 50 images were split into a training set of 40 images and a test set of 10 images. The split was fixed using a seeded random shuffle (seed=42) to ensure reproducibility, and the identical partition was applied to both models to ensure a fair comparison.

B. Network Architecture

U-Net [2] was the selected segmentation architecture for both models. U-Net was specifically designed to train end-to-end from very few images, using an encoder-decoder structure with skip connections that link corresponding layers in the contracting and expanding paths. The skip connections pass high-resolution spatial features from the encoder directly to the decoder, recovering fine boundary detail that would otherwise be lost during downsampling.

The encoder depth was set to 3, producing a bottleneck resolution of 32×40. A shallower encoder reduces the number of trainable parameters, directly lowering overfitting risk on a 35-image training set. A depth of 4 was considered but rejected as the resulting 16×20 bottleneck was too far a compression for detecting minority-class objects occupying as little as 1.6% of pixels. Both models use an identical architecture to isolate training strategy as the sole independent variable in the comparison.

To address the class imbalance, a weighted cross-entropy loss function was used. Weighted cross-entropy assigns higher

loss penalties to underrepresented classes by weighting each class inversely proportional to its pixel frequency, balancing the influence of each class during training [3]. The loss is computed as:

$$L = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C w_c y_{i,c} \log(p_{i,c}) \quad (1)$$

where N is the total number of pixels, C is the number of classes, w_c is the weight for class c , $y_{i,c}$ is the ground truth label, and $p_{i,c}$ is the predicted probability for pixel i belonging to class c .

The baseline model uses weights derived from pixel frequency across all 50 images, normalised relative to the background: background 1.0, weeds 24.66, crops 58.07. The improved model uses modified weights informed by confusion matrix analysis of an initial pilot training run. This is described further in Section II-D.

The Adam optimiser [4] was used for all training runs. It was chosen for its adaptive per-weight learning rate and computational efficiency.

C. Baseline Model

The baseline model was trained on the 40 original training images without any data augmentation. It serves as a performance reference, establishing the segmentation quality under standard training conditions. Since architecture, loss function, optimiser, and data split are identical between both models, any performance difference can be directly attributed to the training strategies of the improved model.

Training was conducted for 15 epochs with a mini-batch size of 8, an initial learning rate of 1×10^{-3} decaying by a factor of 0.1 every 8 epochs, with data shuffled between every epoch. Final evaluation was performed on the test set of 10 images.

D. Improved Model

The improved model uses the same U-Net architecture as the baseline, differing only in training strategy. Development followed an iterative process involving a pilot run and a refined final run. The pilot run applied two strategies to address data scarcity and class imbalance: Data augmentation expanded the training set through spatial transforms and colour jitter. Class-focused patch extraction oversampled minority class regions by extracting patches centred on crop and weed pixels. The loss function used the original baseline class weights normalised relative to background (background=1, weeds=24.66, crops=58.07), and training was performed on 35 images with a validation set of 5 images for checkpoint selection.

Analysis of the confusion matrix and segmentation metrics of the pilot run revealed two key failure modes. First, 28.9% of crop pixels were misclassified as weeds, indicating the high weed weight was creating too strong a pull toward weed predictions. Second, weed IoU was consistently poor across test images since the network had insufficient exposure to large weed regions.

The final model incorporated these changes:

- **Modified class weights:** background weight was increased from 1 to 5 to penalise weed-background misclassification, weed weight reduced from 24.66 to 20 to reduce crop-weed confusion, crop weight remained unchanged.
- **Class-specific patch scaling factors:** were adjusted to increase weed and crop patch counts: background scaling factor 20, weeds reduced to 10, crops set to 12 (described further in section II-F).
- **Larger patches** (800×960) for weed-dominant images were extracted to expose the network to full weed regions.
- **Weed-dominant image oversampling:** images with weed content above 10% receive up to $3\times$ more augmented versions, directly addressing the scarcity of large weed region examples (refer section II-E for further details).
- **Removal of the validation set**

These refinements, combined with the augmentation and patch extraction strategies described in Sections II-E and II-F, expanded the effective training set from 40 to 468 samples. Training was conducted for 20 epochs with a mini-batch size of 8, an initial learning rate of 1×10^{-3} decaying by a factor of 0.1 every 7 epochs. Final evaluation was performed on the same 10-image test set as the baseline.

E. Data Augmentation

The augmentation pipeline described here is that of the final improved model. In the pilot run, each training image generated 3 augmented versions with a uniform augmentation rate. The final run introduced weed-dominant image oversampling. Each training image generates 3 augmented versions by default, with weed-dominant images receiving up to 9. Augmentations were applied to the training set only. The augmentations were of 2 types:

- **Spatial Transformation:**
 - Horizontal flips (50 % probability)
 - Vertical flips (50% probability)
 - Random rotations 45 with geometric safe crop (50% probability)
- **Colour Transformations:**
 - Brightness adjustments (20%)
 - Contrast increase (0 – 30%)
 - Saturation increase (0 – 30%)
 - Gaussian noise ($\sigma = 0.02$)

All spatial augmentations were applied at full resolution before resizing to preserve detail prior to transformation. Pixel values were normalised to [0,1] after each colour augmentation. The rotation used a geometric safe-crop to prevent the network from learning empty corner regions as background features. The largest valid interior rectangle after rotation was computed as:

$$W_{new} = \frac{W_{orig}}{\cos(\theta) + \sin(\theta) \cdot \frac{W_{orig}}{H_{orig}}} \quad (2)$$

with H_{new} derived proportionally to preserve the original aspect ratio. Additionally, rotational angles close to 0, 90, 180, and 270 were excluded to avoid generating near duplicates of the flipped images.

Colour augmentations were applied to the image only since pixel class labels are independent of photometric properties. It ensures the model learns semantic features rather than overfitting to specific lighting conditions of the training data [5]. Only increases in contrast and saturation were applied as the training images already exhibit low values in both.

F. Class-Focused Patch Extraction

Crops and weeds occupy only 1.6% and 3.8% of total pixels respectively. Despite the weighted loss function, the network sees relatively few minority class pixels per training iteration. Patch extraction directly addresses this by oversampling image regions containing underrepresented classes [6].

Patches were extracted from full-resolution images before resizing, to preserve details such as leaf/weed edges before downsampling. Each patch was subsequently resized to 256×320 for network input. The pilot run used a uniform patch size of 640×480 for all images. In the final run, weed-dominant images used larger patches of 800×960 to capture full weed regions.

For each training image, patch centres were sampled from pixel locations belonging to each class, with counts proportional to the class weights computed in section II-B:

$$n_c = \max \left(1, \text{round} \left(\frac{w_c}{s_c} \right) \right) \quad (3)$$

where n_c is the number of patches extracted for class c , w_c is the corresponding class weight, and s_c is a class-specific scaling factor. The pilot run used a uniform scaling factor of 20 for all classes. In the final run, scaling factors were adjusted based on confusion matrix analysis: background $s=20$, weeds $s=10$ (increased weed patch count), crops $s=12$ (increased crop patch count). This yielded approximately 6 weed patches, 5 crop patches, and 1 background patch per image for the final run.

Each extracted patch was augmented with the spatial and colour transformations, as described in Section II-E. Rotation was not applied to patches as the safe-crop would have further reduced the size of the already small patch.

G. Training

Both models were trained using the `trainnet` function in MATLAB R2025b. The baseline was trained on the 40 original images for 15 epochs. The improved model was trained on 468 samples for 20 epochs. Both used a mini-batch size of 8, an initial learning rate of 1×10^{-3} with piecewise decay by a factor of 0.1, and training data shuffled each epoch. The learning rate drop period was set to 8 epochs for the baseline and 7 epochs for the improved model to provide evenly spaced drops.

TABLE I
PERFORMANCE COMPARISON BETWEEN MODELS.

Metric	Baseline	Improved	Change
IoU			
Background	0.9656	0.9835	+0.018
Weeds	0.1828	0.6154	+0.433 (+237%)
Crops	0.3946	0.7551	+0.361 (+91%)
Mean	0.5143	0.7846	+0.270 (+53%)
BFScore			
Background	0.8775	0.9489	+0.071
Weeds	0.5513	0.6887	+0.137
Crops	0.6698	0.9105	+0.241 (+36%)
Mean	0.6995	0.8494	+0.150 (+21%)
Precision			
Background	0.9973	0.9952	-0.002
Weeds	0.2711	0.6954	+0.424 (+156%)
Crops	0.4123	0.8446	+0.432 (+105%)
Mean	0.5602	0.7725	+0.212 (+38%)
Recall			
Background	0.9681	0.9881	+0.020
Weeds	0.3594	0.8424	+0.483 (+134%)
Crops	0.9022	0.8769	-0.025
Mean	0.7433	0.9025	+0.159 (+21%)
F1			
Background	0.9825	0.9917	+0.009
Weeds	0.3091	0.7619	+0.453 (+147%)
Crops	0.5659	0.8604	+0.295 (+52%)
Mean	0.6192	0.8713	+0.252 (+41%)
Accuracy	0.9490	0.9821	+0.033

III. EVALUATION

To test the performance of both models, the following metrics were measured:

- **Intersection over Union (IoU)**: measures the overlap between predicted and ground truth regions for each class.
- **Boundary F1 Score (BFScore)**: the accuracy of predicted segmentation boundaries against ground truth. Focusing on edge quality rather than region overlap.
- **Precision**: fraction of predicted pixels of a class that are correct
- **Recall**: fraction of true pixels of a class that were detected.
- **F1**: harmonic mean of precision and recall, penalising large imbalances between the 2.

Table I shows the per-class and mean performance of both models across all evaluation metrics. Overall pixel accuracy is reported as a secondary metric only. With background constituting 94.5% of all pixels, a model predicting background everywhere would achieve 94.5% accuracy without detecting a single plant. IoU and F1 are therefore more informative as they explicitly penalise missed detections and false alarms per class independently.

The improved model shows substantial gains across all minority class metrics, most notably weed IoU increasing from 0.18 to 0.62 and crop IoU from 0.39 to 0.76. The confusion matrices (shown in Table II) provide insight into where these gains came from. In the baseline, 55.9% of weed pixels were

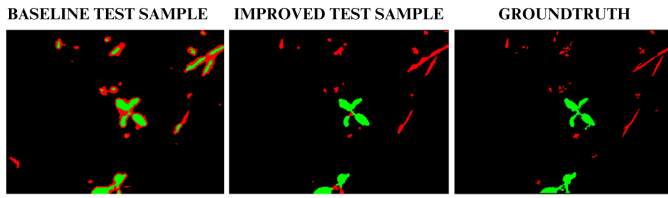


Fig. 1. Segmentation comparison between the baseline and improved models.

TABLE II
CONFUSION MATRICES FOR THE BASELINE AND IMPROVED MODELS

Baseline				Improved			
T\P	Bg	Wd	Cr	T\P	Bg	Wd	Cr
Bg	96.8	2.8	0.3	Bg	98.8	1.0	0.2
Wd	8.2	35.9	55.9	Wd	13.7	84.2	2.1
Cr	0.3	9.5	90.2	Cr	2.9	9.4	87.7

predicted as crops. The network had seen very few large weed regions during training and mistook them for crops. The improved model reduced this to 2.1%, which can be attributed to weed-dominant image oversampling and larger patch sizes for weed-heavy images, both of which increased the network’s exposure to large weed regions. The weed-background confusion worsened slightly from 8.2% to 13.7%, a consequence of reducing the weed class weight to fix the crop-weed confusion from the pilot run. Another possible reason for this is that resizing the image loses fine detail, and smaller weed areas are lost. Crop performance improved across all metrics, with precision rising from 0.41 to 0.84, consistent with the increased crop patch count giving the network more crop pixel examples per training iteration. Background performance remained considerably stable, confirming that improvements to minority classes did not come at the cost of majority class accuracy. The slight drop in crop recall from 0.90 to 0.88 reflects the network becoming more selective about crop predictions, which is an acceptable trade-off given the large precision gain.

Fig. 1 shows a three panel comparison of baseline prediction, improved prediction, and ground truth for a representative test image. In the baseline prediction, weed regions are misclassified as crops, and crop boundaries are incorrectly labelled as weeds. The improved model corrects the weed regions and crop boundaries, however slight misclassification of crops as weeds remains.

A. Limitations

The iterative approach of analysing the pilot run before finalising the improved model was effective. The improved model performed substantially better than the baseline model, although it still has some limitations. The 10-image test set is too small to draw statistically reliable conclusions. Metric values are sensitive to which images happen to fall in the test set, and a different random split would likely produce different results. With only 6 images across the entire dataset containing

weed content above 10%, the training data is insufficient for precise weed detection regardless of the augmentation. This explains why significant weed-background confusion persisted even in the final model.

Resizing images from 1296×966 to 256×320 loses fine detail. Thin weed stems and small plant boundaries are likely partially responsible for the boundary misclassification still visible in the improved model. A sliding window inference approach at full resolution would improve boundary quality but was not feasible given the CPU training constraints of this project. The small batch size and limited epochs which were also imposed by CPU constraint meant that the improved model hasn’t fully converged.

B. Future Directions

The most direct improvement would be collecting more labelled data, particularly those with high weed content and crop content. However, the dataset is unlikely to grow significantly due to the cost of manual annotation. Therefore future improvements should focus on maximising the utility of existing data with better networks and training methods. Synthetic data generation using generative adversarial networks (GANs) [7] trained on the existing 50 images could produce realistic additional training samples. Diffusion models offer a more recent alternative for generating photorealistic agricultural imagery with controllable class distributions [8]. Semi-supervised learning offers another direction. A small labelled set like this one could be combined with a larger pool of unlabelled field images to generate pseudo-labels for unlabelled images. This would require no additional manual annotation and would expand the dataset.

Future directions should also consider the applications of such weed detection models. The most practical deployment is integration with drone-based systems. A drone equipped with a downward-facing camera could capture field images at regular intervals with GPS coordinates and timestamps, enabling precise and targeted herbicide application. However, U-Nets are too slow for drone application. BiSeNet v2 would be a more suitable network as its dual-pathway architecture separates spatial detail from semantic context, combining them through a lightweight feature fusion module designed for real-time inference on limited hardware [9].

IV. CONCLUSION

This paper trained two U-Net segmentation models on a 50-image subset of the SugarBeets 2016 dataset for crop and weed detection. The baseline model established a reference performance under standard training conditions, while the improved model addressed the core challenges of data scarcity and class imbalance through data augmentation, class-focused patch extraction. The improved model achieved a mean IoU of 0.78 compared to 0.51 for the baseline, with weed IoU improving by 237% and crop IoU by 91%. The iterative development process proved effective and produced consistent improvements across all minority class metrics. The primary

limitation remains the dataset size. With only 50 images, reported metrics are sensitive to the train/test split.

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